DABIN KIM

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Education

Hanbat National University (Advisor: Prof. Haneol Jang) *Daejeon*

B.S. in Department of Computer Engineering Mar. 2022 - 2026

GPA 4.14/4.5 (Major: 4.2/4.5)

Hanbat National University (Advisor: Prof. Haneol Jang) *Daejeon*

M.S. in Department of Computer Engineering Mar. 2026 - Present

GPA 4.5/4.5

Publications

Conference

[1] **Dabin Kim**, Seoyeon Oh, Dongsu Kim, Cheoneum Park, and Haneol Jang, "Face anti-spoofing method using contrastive learning and color jittering", *KDFS2024*, May 2024. **(Best Paper Award)**

[2] **Dabin Kim**, Woojin Park, Eunkyung Kim, and Haneol Jang, "Enhancing Zero-Shot Detection of Military Objects using Vision-Language Models via Prompt Tuning", *KIBME2025*, May 2025.

[3] Woojin Park, **Dabin Kim**, Eunkyung Kim, and Haneol Jang, "Research on Military Vehicle Object Detection under Data Scarcity using Diffusion Model-based Background Reconstruction Synthetic Data", *KIBME2025*, May 2025.

[4] **Dabin Kim**, Taehoon Kim, and Haneol Jang, "Enhancing SAR Detection via Auxiliary Angle-Loss Learning", *KIBME2025 Winter Conference*, Dec. 2025. **(Encouragement Award)**

Journal

[1] **Dabin Kim**, Seo-Yeon Oh, Dong-Su Kim, Cheoneum Park, and Haneol Jang, "Face anti-spoofing method using contrastive learning and color jittering", *Journal of Digital Forensics*, vol. 18, no. 5, pp. 40–49, Dec. 2024.

[2] **Dabin Kim**, Woojin Park, Eunkyung Kim, and Haneol Jang, "Enhancing Zero-shot Detection of Military Object using Vision-Language Models via Prompt Tuning", *Journal of IKEEE*, vol. 29, no. 3, pp. 297–303, Sep. 2025.

[3] Woojin Park, **Dabin Kim**, Eunkyung Kim, and Haneol Jang, "Research on Military Vehicle Object Detection using Diffusion Model-based Background Reconstruction Synthetic Data", *Journal of IKEEE*, vol. 29, no. 3, pp. 331–340, Sep. 2025.

[4] **Dabin Kim**, Jihwan Moon, Taehoon Kim, and Haneol Jang, "TA-OSDet: Text Fusion for Optical-SAR Rotated Object Detection", *Journal of IKEEE*, vol. 30, no. 1, pp. 49–56, Mar. 2026.

[5] **Dabin Kim**, Seongmin Pyo, Seoyeon Oh, Kampol Woradit, Haneol Jang (co-corresponding author), and Dong-Hyo Lee, "QueryFieldNet: A Query-Conditioned Surrogate Model for Multi-Beam Reflectarray Principal-Cut Gain Prediction", **Mathematics (SCIE, IF 2.3, Top JCR 5.7%)**, vol. 14, no. 17, article 3080, 2026.

[6] Seoyeon Oh, **Dabin Kim**, Taehoon Kim, Jihwan Moon, and Haneol Jang, "Tri-Harmony-NCD: Multi-View Relations and Trust-Gated Graph Decoding for Novel Class Discovery of Industrial Anomalies", **Scientific Reports (SCIE, IF 4.9)**, **under review.**

Research Experience

AIM Lab., Hanbat National University *Daejeon, Republic of Korea*

B.S. Researcher (Advisor: Prof. Haneol Jang) May 2022 - Feb. 2026

- Conducted AI and computer vision research on face anti-spoofing, **zero-shot military object detection**, and SAR object detection

Datamaker *Daejeon, Republic of Korea*

Internship Aug. 2025 - Oct. 2025

- Developed **zero-shot object detection for military applications**, collected new data, and proposed methods to improve detection performance

AIM Lab., Hanbat National University *Daejeon, Republic of Korea*

M.S. Researcher (Advisor: Prof. Haneol Jang) Mar. 2026 - Present

- Researching **single-image building damage assessment** with vision foundation models and deep learning-based antenna gain prediction

Research Projects

Face Anti-Spoofing for Image-Based Authentication *Python, PyTorch*

AI Course Team Project Mar. 2024 - Jun. 2024

- Developed data augmentation and deep learning methods for **detecting face presentation attacks**

Research on Lightweight Image Steganalysis *Python, PyTorch*

National Security Research Institute Aug. 2024 - Sep. 2024

- Generated original and information-embedded image pairs and **implemented and evaluated image steganalysis methods**

Deep Learning for X-ray Diagnosis of Neonatal Respiratory Distress Syndrome (RDS) *Python, PyTorch*

Chungnam National University Hospital Oct. 2024 - Dec. 2024

- Preprocessed chest X-rays, applied lung segmentation, and trained and evaluated deep learning classifiers

Zero-Shot Object Detection in Aerial Imagery *Python, PyTorch*

Capstone Design Jan. 2025 - Feb. 2026

- **Developed and refined zero-shot military object detection methods** using multimodal learning and prompt engineering
- Collected and preprocessed aerial imagery; augmented training data with synthetic images to address data scarcity

Building Damage Assessment from Remote Sensing Imagery *Python, PyTorch*

AIM Lab., Hanbat National University Mar. 2026 - Jun. 2026

- Investigated **deep learning-based building damage assessment** using remote sensing imagery

Chat AI for Cybersecurity Policy Knowledge *Python*

National Security Research Institute 2026 - Present

- Developing **evidence-grounded question answering** over cybersecurity policy documents using retrieval-augmented generation
- Implementing document preprocessing, metadata-aware search, and **traceable source citations** to support reliable responses

AI-Based Multi-Beam Reflectarray Antenna Gain Prediction *Python, PyTorch*

Research Project, AIM Lab., Hanbat National University Jun. 2026 - Aug. 2026

- Developed and evaluated **a deep learning surrogate for antenna gain prediction** from phase configurations and angular queries

Single-Image Building Damage Assessment *Python, PyTorch*

AIM Lab., Hanbat National University Sep. 2026 - Present

- Developing **building localization and damage classification from a single post-disaster image**
- Evaluating region-level predictions and training strategies through controlled comparative experiments

Awards and Honors

1st Place	Image classification (2024 spring), Hanbat National University (Undergraduate, Kaggle)	*Spring 2024*
Best Paper Award	Korean Digital Forensics Society (KDFS2024)	*Jun. 2024*
Silver Prize	SW-univ AI Competition 2024 (Fake Audio Detection)	*Jun. 2024*
Gold Prize	The 14th 2025 Computer Engineering Capstone Design	*Dec. 2025*
Encouragement Award	KIBME 2025 Winter Conference Competition	*Dec. 2025*

Technical Skills

Programming	Python, Java, C, C++
Frameworks	PyTorch
Languages	Korean (Native), English (Intermediate)

Teaching Experiences

Artificial Intelligence Class *Hanbat National University*

Teaching Assistant (Prof. Haneol Jang) Mar.–Jun. 2026

Computer Vision Class *Hanbat National University*

Teaching Assistant (Prof. Haneol Jang) Sep.–Dec. 2026